TEXORA · LECTURE SHEET

# Chemical Finishing of Garments

*Functional finishes applied at garment stage — softening, wrinkle-free resin, water repellency, flame retardancy and antimicrobial treatment — their chemistries, application routes and dosing calculations.*

AM-4101 · ADVANCED APPAREL MANUFACTURING-I · TOPIC 6 · SPECIALIZATION — APPAREL MANUFACTURING

## 1. Finishing the garment, not the cloth

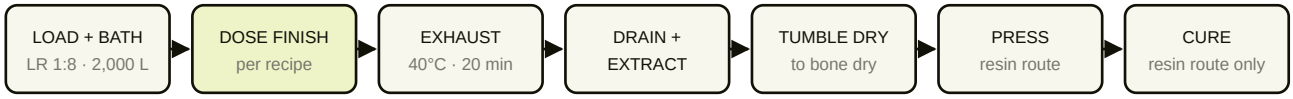
Most chemical finishing is done on fabric, but a defined group of finishes is applied to the **sewn garment**: when the effect must cover seams uniformly (softening, fragrance and antimicrobial treatments in the wash plant), when the fashion product is a treated garment (wrinkle-free shirts post-cured after pressing), or when only sold garments should carry the cost (buyers order finishing on confirmed styles). Garment application uses the wet-processing plant of Topics 1–2 — exhaust application in washing machines, spray application in booths, and pad application for piece goods analogues — followed by drying and, for resin systems, curing.

## 2. The principal garment finishes

| FINISH | CHEMISTRY | EFFECT AND NOTES |
|---|---|---|
| Softening | Cationic softeners; silicone (amino-functional) softeners for premium hand | Softer, smoother hand; silicones add surface smoothness and slight sewability gain. Exhaust-applied in the last bath |
| Wrinkle-free / easy-care | Crosslinking resin (DMDHEU type) + $MgCl_2$ catalyst | Cellulose chains crosslinked so creases recover; garment post-cure route: apply, dry, press, then cure so creases are set sharp and the body stays smooth |
| Water repellent | Fluorocarbon or silicone repellents | Beading and shedding of water without coating the fabric airtight; spray-rated (Topic: comfort testing in AM-4103) |

| | | |
|---|---|---|
| Flame retardant | Phosphorus-based durable systems for cellulosics; topical systems for limited-life uses | Slows ignition and flame spread for workwear and children's sleepwear regulations; durability class must match the claim |
| Antimicrobial | Silver-based, quaternary ammonium or natural (e.g. chitosan) agents | Inhibits odour-causing bacteria on the garment; claims regulated — only substantiated wording may be used |
| Bio-polishing | Cellulase enzyme (the Topic 1 enzyme, used for surface quality) | Removes surface fuzz: cleaner colour, less pilling; a finishing effect from a washing chemical |
| Easy-care / moisture management | Hydrophilic finishes on synthetics | Improves wicking and soil release on polyester garments |

## 3. Application routes




Exhaust finishing sequence in the washer-extractor; the press-and-cure steps apply to the wrinkle-free resin route

Figure 1 — Garment finishing by exhaust application. Softener stops after drying; resin continues through pressing to curing, so garment creases are crosslinked in their pressed shape.

## 4. Dosing calculations

**WORKED EXAMPLE — A 250 KG FINISHING BATCH**

Batch: **250 kg** of garments, liquor ratio 1 : 8 → bath = 250 × 8 = **2,000 L**.

**Softener** at 2% owg = 250 × 0.02 = **5.0 kg**. (The same finish dosed on bath volume at 3 g/L would be 2,000 × 3 = 6.0 kg — the two conventions differ; the recipe states one, see Topic 1.)

**Wrinkle-free resin (DMDHEU)** at 8% owg = 250 × 0.08 = **20 kg**.

**Catalyst ($MgCl_2$)** at 20% on the weight of resin = 20 × 0.20 = **4.0 kg** — catalysts are dosed on the resin, a third basis; all three bases appear on one batch sheet, each labelled.

## 5. Trade-offs and testing

Finishes are compromises measured by test. Resin crosslinking buys crease recovery at the price of tensile and tear strength (the crosslinks that stop chains moving also stop them sharing load) and, in

older chemistries, formaldehyde release — modern low-formaldehyde resins and correct curing keep garments within buyer limits, verified by strength and formaldehyde testing. Softener improves hand but can reduce water absorbency and, in excess, cause seam slippage complaints. Every finish claim on a worksheet — repellency rating, antimicrobial efficacy, flame performance — must be verified by the corresponding standard test (AM-4103 covers comfort and protective testing) before it is printed on a label: the finish is not finished until it is measured.

## Glossary

| TERM | MEANING |
|---|---|
| DMDHEU | The standard cellulose crosslinking resin for wrinkle-free finishes. |
| Catalyst | Agent (here $MgCl_2$) that enables resin crosslinking during curing; dosed on resin weight. |
| Post-cure | Garment route in which resin is dried into the garment, the garment is pressed, then cured to set the pressed shape. |
| Exhaust application | Finish taken up by the garments from a bath in a washing/dyeing machine. |
| Cationic softener | Positively charged softener substantive to (negatively charged) cellulose, applied from the final bath. |

## Key points and summary

- Garment-stage finishes: softening, wrinkle-free resin, repellency, flame retardancy, antimicrobial and bio-polishing — applied by exhaust, spray or pad routes in the wet plant.
- The wrinkle-free post-cure route (apply → dry → press → cure) sets creases by crosslinking them in place.
- One batch sheet can carry three dosing bases — % owg (softener 5.0 kg, resin 20 kg), g/L, and % on resin (catalyst 4.0 kg) — each must be labelled.
- Every finish trades against something (strength, absorbency) and is only real when verified by its standard test.

## Practice

Answer in your own words; brief model answers follow.

1. Scale the worked dosing to a 180 kg batch at the same ratios: softener at 2% owg, resin at 8% owg, catalyst at 20% on resin.

2. Why is the catalyst dosed as a percentage of the resin rather than of the garments or the bath?
3. Explain why resin finishing reduces tear strength although it improves crease recovery.
4. A buyer requests an antimicrobial claim on the care label after a finish is applied. What must happen before the claim is printed?

## Answer key

1. Softener = 180 × 0.02 = 3.6 kg. Resin = 180 × 0.08 = 14.4 kg. Catalyst = 14.4 × 0.20 = 2.88 kg.
2. The catalyst's job is to drive the resin's crosslinking reaction, so its required amount scales with the amount of resin present — not with garment mass or water volume, which are incidental to that chemistry.
3. Crosslinks lock cellulose chains together: creases recover because chains cannot slide into new positions — but for the same reason a tearing load cannot be shared by chains sliding and grouping, so the fabric breaks at lower force.
4. The finish's efficacy must be verified by the applicable standard antimicrobial test on the finished garment (including after the claimed number of washes), and the claim wording checked against the buyer's and market's rules. An applied chemical is not yet a substantiated claim.

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